

SERS Dataset Acquisition Protocol

Substrate-agnostic chemical classification dataset

Goal. Build a complete chemical-by-substrate matrix/dataset.

Core matrix.

Item	Target
Substrate families	Ag, Au, PICO, pSERS
Chemicals	Ideally 8 chemicals that respond on all 4 substrates; minimum 5 if time-limited
Chemical-substrate pairs	Ideal: $8 \times 4 = 32$ pairs; minimum: $5 \times 4 = 20$ pairs
Spectra per pair	Ideal: 200+ usable spectra; minimum pilot: 100 usable spectra
Ideal analyte total	$32 \times 200+ = 6400+$ spectra
Minimum pilot total	$20 \times 100 = 2000$ spectra
Blanks/controls	Per substrate family: 100+ substrate-only spectra

Per chemical-substrate pair: collect 200+ usable spectra across independent samples.

Timing	Independent sample	Substrate piece/batch	Maps	Spectra
Preferred day 1	Preparation A	Piece/batch 1	2 maps x 25	50
Preferred day 1	Preparation B	Piece/batch 2	2 maps x 25	50
Preferred day 2	Preparation C	Piece/batch 1 or 3	2 maps x 25	50
Preferred day 2	Preparation D	Piece/batch 2 or 4	2 maps x 25	50
Total per chemical-substrate pair				200

Procedure.

1. For each chemical-substrate pair, prepare four independent samples: A, B, C, and D.
2. Collect two 5 x 5 maps from each preparation. Each map gives 25 spectra.
3. Prefer acquisition over two days. Same-day acquisition is perfectly fine if the samples are independently prepared and measured on different substrate pieces/batches.
4. What we don't want: all spectra from one preparation, one substrate piece, one spot, or one map.
5. For every substrate family, collect 100 substrate-only spectra.

Note. If more independent samples, substrate pieces/batches, acquisition days, or spectra can be prepared, that is better. More independent variation will make the substrate-agnostic dataset stronger.